

Chairman of Isro, told his colleagues at the agency's Mission Operations Complex in Bengaluru. He should know, for he was around when the lander sent with the Chandrayaan-2 mission crashed on the moon's surface in September 2019. For nearly four years thereafter, he and the rest of the team worked day and night to perfect the third mission.

In its life span of one lunar day, equivalent to 14 earth days, the Vikram Lander, Pragyan Rover and the propulsion module have all commenced a series of scientific experiments. The four payloads on Vikram Lander will measure the density of changes in near-surface plasma comprising ions and electrons, thermal properties of the lunar surface near polar regions, seismicity around the landing site and delineating the structure of the lunar crust and mantle, as well as conduct a passive experiment to understand the dynamics of the moon system. To learn more about the lunar surface, the payloads on Pragyan (or 'wisdom') Rover will conduct qualitative and quantitative elemental analysis.

Experts believe Chandrayaan-3 will accelerate India's self-reliance in emerging sectors and tech and boost its confidence. "Chandrayaan-3 is a classic lesson on focussed perseverance, economic innovation and sovereignty in new and emerging tech. The event will propel India's space economy," says Sreeram Ananthasayanam, Partner at Deloitte India.

INDIA INC.'S CONTRIBUTION

Another crucial aspect here is the role played by India Inc. in developing essential technologies and mission-related components. Experts see this as yet another milestone in the evolution of the 400-odd private companies that form a part of the extensive vendor ecosystem that Isro has diligently developed over the last 54 years. Lt Gen. (Retd) Anil Kumar Bhatt, Director General of the apex industry association, Indian

CHANDRAYAAN-3: TO THE DARK SIDE OF THE MOON

The Chandrayaan-3 probe has landed near the lunar South Pole, a region with water ice or frozen water, which could serve as a source of oxygen, fuel and water for future missions

MISSION COST

\$75 million

LAUNCH DATE

July 14

LANDING DATE

August 23

DIMENSIONS

Vikram Lander

**Height - 2 metres;
Mass - 1,700 kg**

Pragyan Rover

Mass - 26 kg

LIFECYCLE

**One lunar day
(equivalent
to two weeks on
earth), during which
time the lander,
rover and propulsion
module will conduct
a series of scientific
experiments**

Space Association (ISpA), believes this will lead to the development of a thriving space economy. "An important step forward in space exploration and commercialisation will be with the inclusion of more private players and our growing spacetechn start-ups in the future missions to make India a leader in the global space economy," he says.

For Chandrayaan-3, Larsen & Toubro made launch vehicle booster segments and subsystems, Bharat Heavy Electricals supplied batteries, Kerala State Electronics Development Corp. developed electronic power modules and test and evaluation systems, and Walchand Industries made mission components. Subba Rao Pavuluri, Chairman and MD of Hyderabad-based Ananth Technologies, which supplied several mission-critical systems for the mission, says, "Chandrayaan-3 provides opportunities for co-operation with other space agencies, researchers, and institutions worldwide."

Similarly, Godrej Aerospace, a subsidiary of Godrej & Boyce, provided liquid propulsion engines, satellite thrusters and control module components for the Chandrayaan and Mangalyaan missions. "This success reinforces our belief in the power of indigenous manufacturing and our contribution to pushing the boundaries of technological innovation," says Jamshyd Godrej, Chairman & MD of Godrej & Boyce.

Amit Sharma, MD & CEO of Tata Consulting Engineers, a subsidiary of the Tata group, believes the mission will show India's potential and scientific pioneering. "The success of this mission will inspire young scientific minds and future scientists and engineers, enabling India to innovate across sectors and take the Make in India mission to new heights."

Chandrayaan-3 has, thus, literally taken India's space sector over the moon. **BT**

@manishpant22